

Chapter

Reflection

1. Use graph sheet to answer this question. Take $2\text{ cm} = 1\text{ unit}$ along both the axes.

(i) Plot A, B, C where $A(0, 4)$, $B(1, 1)$ and $C(4, 0)$

(ii) Reflect A and B on the x -axis and name them as E and D respectively.

(iii) Reflect B through the origin and name it F . Write down the coordinates of F .

(iv) Reflect B and C on the y -axis and name them as H and G respectively.

(v) Join points A, B, C, D, E, F, G, H and A in order and name the closed figure formed.

Main 2023

Solution

1.

Ans: Take 2 cm = 1 unit on both the axes.

(i) The points $A(0, 4)$, $B(1, 1)$ and $C(4, 0)$ are plotted on the graph as shown.

(ii) Reflecting points $A(0, 4)$ and $B(1, 1)$ in the x -axis, we get

$$E(0, -4) \text{ and } D(1, -1)$$

as shown in the figure.

(iii) Reflecting point $B(1, 1)$ through the origin, we get

$$F(-1, -1)$$

as shown in the figure.

(iv) Reflecting points $B(1, 1)$ and $C(4, 0)$ in the y -axis, we get

$$H(-1, 1) \text{ and } G(-4, 0)$$

as shown in the figure.

(v) Joining the points

$$A, B, C, D, E, F, G, H \text{ and } A$$

in order, the closed figure obtained is a **star**.

